

# Context Window

When you type  $2 + 2$  on the calculator on your phone, you always get 4. Same input, same answer, every time. AI doesn't work that way. Consider the following: identical questions, asked to AI by two different people.

**Luke**

*Asks AI*

What car should I buy after I graduate from college?

*ChatGPT answers*

Great question, Luke. I definitely recommend a **Jeep Cherokee**.

**Nate**

*Asks AI*

What car should I buy after I graduate from college?

*ChatGPT answers*

I'm happy you asked this question, Nate. At this point in your life, I recommend the **Ford Raptor**.

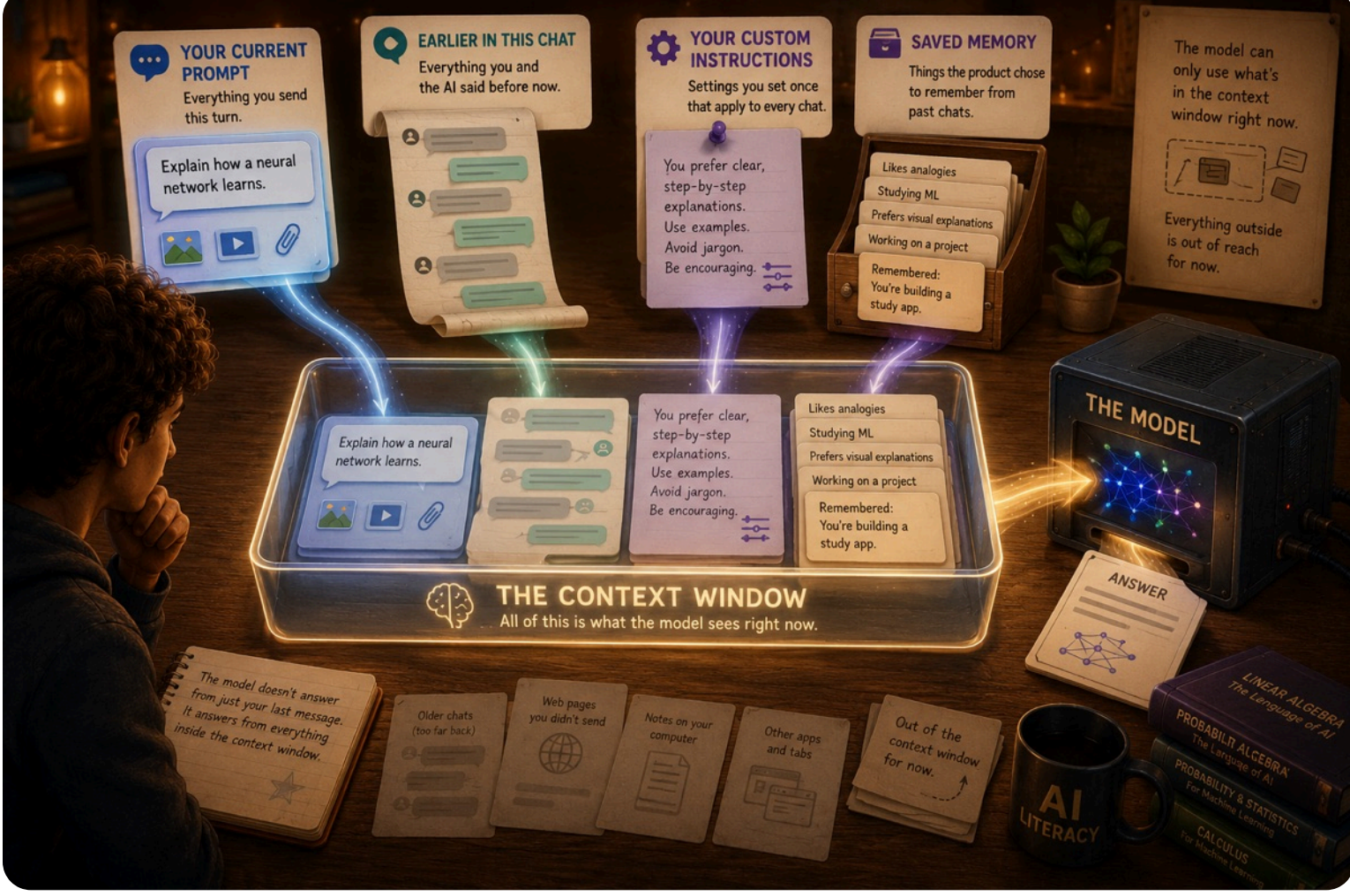
Wait a second. In the last lab, you changed the answer by changing the prompt. Luke and Nate typed the exact same prompt into the same app, and still got different answers!

Some people think this is a reason not to trust AI. But the opposite is true. It's actually demonstrating the power of AI. It gave a different, more tailored answer to each.

## WHY DID THIS HAPPEN?

When you use a specific model of ChatGPT, it's the same tool everyone else starts with. But ChatGPT knows more than just the last message Luke and Nate typed. For example, earlier in his chat, Nate said he loves pickup trucks. When ChatGPT answers his question, it recognizes this and suggests the Ford Raptor.

What do you call everything the model can see when it answers your question? The **context window**. Think of it almost like the working memory the model uses to answer. And it builds that working memory from several places.



## WHAT THE MODEL ACTUALLY SEES

Here's the part that surprises people. When you hit send, the model doesn't just see your last message. It sees the entire context window, and it answers from all of it. The illustration above shows everything that feeds it: your prompt, the chat so far, your custom instructions, and saved memory.

## MAKE IT YOURS

The two feeds beyond the chat are yours to set up, and every major app has some version of them. They usually come in three flavors:

**Personalization**

Tell it once and it applies to every chat: what to call you, what you're into, how you like answers.

**Projects**

Instructions and files scoped to one piece of work, like one class or one essay.

**Memory**

Things the app picks up from your chats and saves on its own.

One thing to keep straight: these are app features, not the model learning about you. The app stores what you set, then quietly adds it to the context window every time you hit send.

## CONTEXT WINDOW SIZE

The whole window is measured in tokens, the same chunks you met in the Tokens lesson. Pile up enough of them and you hit the context window's limit.

Before that limit, the model sees everything in the chat. After it, it doesn't. Some apps summarize the older parts. Others just forget.

And that limit is growing fast.

**4,096 TOKENS**

**1st version of ChatGPT (November, 2022)**

~3,000 words — About 6 pages

**1 MILLION TOKENS**

**Latest Claude models**

~1,000,000 words — About the size of the first 5 Harry Potter novels

Big or small, what fills the window is what shapes the answer. Same question, different windows.

